**Koneru Lakshmaiah Education Foundation**
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# STUDENT CASE STUDY EXECUTION REPORT

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|---|---|
| Department | FED |
| Subject | DSA-2 |
| Academic Year | I-III Semester (2025-26) |
| Faculty Name | Dr Sukanya Ledalla |
| Student Name | Billakanti Siri |
| Roll Number | 2520030601 |
| Branch | CSE |
| Course Outcome | CO6 – Coin Change Dynamic Programming |

**Case Study Topic:** Coin Change for ₹ Denominations using Dynamic Programming

## Work Done Summary:

In this case study, we implemented the Coin Change Dynamic Programming algorithm to determine the minimum number of coins required to make a given amount. The problem uses Indian coin denominations {1, 2, 5, 10, 20} and finds the optimal solution for amount ₹43. Dynamic Programming was used to guarantee the minimum coin count and recover the actual coins used.

## Implementation Details:

1.Implemented Coin Change Dynamic Programming using Java.

2.Created a DP table to store minimum coins required for every amount from 0 to 43.

3.Used tabulation (bottom-up DP) approach.

4.Maintained a choice array to recover the coins used.

5.Computed the minimum number of coins required for ₹43.

## Result: Screenshots/output

```
siribillakanti@siris-MacBook-Air VSC DSA II % javac assignment_CO6.java
siribillakanti@siris-MacBook-Air VSC DSA II % java assignment_CO6
Amount = 43
Coins = [1, 2, 5, 10, 20]

Minimum Coins Required = 4
Coins Used = [1, 2, 20, 20]
siribillakanti@siris-MacBook-Air VSC DSA II %
```

**GitHub Link:**
**https://github.com/siribillakanti/DSA2_ASSIGNMENTS/blob/main/assignment_CO6.java**